

CaBP1 and 2 enable sustained $\text{Ca}_V1.3$ calcium currents and synaptic transmission in inner hair cells

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David Oestreicher, Shashank Chepurwar, Kathrin Kusch, Vladan Rankovic, Sangyong Jung, Nicola Strenzke, Tina Pangrsic 

Experimental Otology Group, InnerEarLab, Department of Otolaryngology, University Medical Center Göttingen, 37075 Göttingen, Germany • Auditory Neuroscience Group, Max Planck Institute for Multidisciplinary Sciences, 37075 Göttingen, Germany • Auditory Systems Physiology Group, Institute for Auditory Neuroscience, InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany • Collaborative Research Center 889, University of Göttingen, 37075 Göttingen, Germany • Functional Auditory Genomics, Institute for Auditory Neuroscience, University Medical Center Göttingen, 37075 Göttingen, Germany • Institute for Auditory Neuroscience and InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany • Restorative Cochlear Genomics Group, Auditory Neuroscience and Optogenetics Laboratory, German Primate Center, 37075 Göttingen, Germany • Multiscale Bioimaging Cluster of Excellence (MBExC), University of Göttingen, 37075 Göttingen, Germany

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Abstract

To encode continuous sound stimuli, the inner hair cell (IHC) ribbon synapses utilize calcium-binding proteins (CaBPs), which reduce the inactivation of their $\text{Ca}_V1.3$ calcium channels. Mutations in the *CABP2* gene underlie non-syndromic autosomal recessive hearing loss DFNB93. Besides CaBP2, the structurally related CaBP1 is highly abundant in the IHCs. Here, we investigated how the two CaBPs cooperatively regulate IHC synaptic function. In *Cabp1/2* double-knockout mice, we find strongly enhanced $\text{Ca}_V1.3$ inactivation, slowed recovery from inactivation and impaired sustained exocytosis. Already mild IHC activation further reduces the availability of channels to trigger synaptic transmission and may effectively silence synapses. Spontaneous and sound-evoked responses of spiral ganglion neurons *in vivo* are strikingly reduced and strongly depend on stimulation rates. Transgenic expression of CaBP2 leads to substantial recovery of IHC synaptic function and hearing sensitivity. We conclude that CaBP1 and 2 act together to suppress voltage- and calcium-dependent inactivation of IHC $\text{Ca}_V1.3$ channels in order to support sufficient rate of exocytosis and enable fast, temporally precise and indefatigable sound encoding.

eLife assessment

This **fundamental** study reveals the major role of calcium-binding proteins (CaBP1 and CaBP2) in sustained exocytosis from mouse inner hair cell ribbon synapses. **Compelling** data and analysis from CaBP1/2 double-knockout mice show enhanced calcium channel (CaV1.3) inactivation, slowed recovery from inactivation, and reduced synaptic vesicle exocytosis as assayed by membrane capacitance measurements, as well as greatly reduced in vivo spontaneous and sound-evoked spikes from the postsynaptic spiral ganglion neurons. Importantly, transgenic expression of CaBP2 led to the rescue of hearing capabilities. The continuous transmission of sound-evoked signals from auditory hair cells thus depends on the expression of both CaBP1 and CaBP2 and their suppression of CaV1.3 inactivation.

Introduction

Calcium influx through voltage-gated calcium channels (Ca_vs) is a pivotal initiator of neurotransmission. These channels typically undergo considerable Ca²⁺- and voltage-dependent inactivation (CDI and VDI), which limits the amount of incoming Ca²⁺ ions and thus possible excitotoxicity. In sensory receptor cells such as sensory inner hair cells and photoreceptors, however, the extent of inactivation is significantly reduced to allow encoding of graded and sustained membrane potential fluctuations in response to ongoing sound or visual (t.i. “darkness”) stimuli [reviewed in [1](#)]. The attenuation of channel inactivation may be largely attributed to the action of calcium-binding proteins, CaBPs [2](#), which primarily abate the calmodulin (CaM)-mediated CDI of Ca_v channels [2–7](#). At the inner hair cell (IHC) ribbon synapses, the Ca²⁺ current is carried through the L-type Ca_v1.3 channels [8,9](#). Their biophysical properties may be regulated by several CaBP family members [10,11](#), but most prominently by CaBP2 [12](#). Mutations in the *CaBP2* gene cause moderate to severe autosomal recessive hearing impairment DFNB93 [see e.g. [12–16](#)]. Interestingly, the loss of CaBP2 seems to significantly affect VDI, but not CDI of the IHC Ca²⁺ currents [12](#). Furthermore, isolated disruption of CaBP1 or CaBP4 in mice leads to at most modest enhancement of IHC Ca_v1.3 inactivation [10,11](#). Moreover, the automodulatory C-terminal domain of the long Ca_v1.3 isoform, abundant at the IHC ribbon synapses [17,18](#), does not seem critical for CDI suppression either, as its disruption does not lead to hearing impairment [18](#). The question of which mechanisms are primarily responsible for suppressing IHC Ca_v1.3 CDI thus remains unresolved. Since CaBP2, followed by the structurally similar CaBP1, are the most abundant members of the CaBP family in the cochlea [19](#), we set out to investigate the phenotype of *Cabp1/2* double-knockout (DKO) animals at the systems and cellular levels. Severe hearing impairment that we observed in these animals reveals that CaBP1 and 2 act together to attenuate the inactivation of the IHC Ca_v1.3 channels. Furthermore, we demonstrate that the combined absence of CaBP1 and 2 almost abolishes sustained, and, upon increased hair cell synaptic activity, the fast component of exocytosis. Such phenotype was observed neither in *Cabp1* nor in *Cabp2* single KO animals [11,12](#). To obtain a more comprehensive understanding of the roles of the two calcium-binding proteins, we attempted to rescue the auditory function of the CaBP1/2-depleted auditory system by viral delivery of a *Cabp2* coding sequence in the inner ear. These experiments revealed a substantial recovery of hearing, which may be related to the successful restoration of Ca²⁺ currents and IHC exocytosis upon transgenic expression of CaBP2. Our findings indicate that the functions of CaBP1 and 2 partially overlap at the IHC ribbon synapse. Together, they provide sufficient attenuation of channel inactivation to sustain Ca²⁺ currents and exocytosis required for temporally precise, indefatigable neurotransmission at this synapse.

Results

Pronounced VDI and CDI of Ca_v1.3

channels in CaBP1/2-deficient IHCs

CaBP1 and 2 show high expression levels in the organ of Corti¹⁹, where they may modulate voltage-gated Ca_v1.3 channels at the IHC ribbon synapses (**Fig. 1A**). Deletion of CaBP2 results in an early-onset, progressive hearing impairment in the KO mouse model and moderate to severe hearing impairment, DFNB93, in affected human patients^{12–14}. Loss of CaBP1 in mice, on the other hand, leaves hearing unperturbed in the first few months after birth¹¹. Since these proteins show a high degree of homology², it is plausible that they possess similar features and are partially redundant, at least initially. Therefore, we were interested in understanding their cumulative impact on the function of hair cells and hearing. For this, we analyzed the phenotype of the *Cabp1/2*-DKO animals and tested the extent of functional rescue following the transgenic expression of CaBP2 (**Fig. S1A**) in the ears of the DKO mice.

To investigate possible synergistic functions of the two major IHC CaBPs, we first measured Ca²⁺ currents in the IHCs of 3-4-week-old *Cabp1/2*-DKO and WT animals in the presence of 1.3 mM extracellular [Ca²⁺]. Employing 500 ms-long depolarization steps to maximal Ca²⁺-current potential (-25 to -15 mV) revealed pronounced inactivation of Ca_v1.3 channels in the DKO IHCs (**Fig. 1B**). This was significantly stronger when compared to the inactivation observed in the IHCs of *Cabp1* or *Cabp2* single KOs in similar conditions (**Fig. 1C**)^{11,12,20}, and even when comparing to data from 5-week-old *Cabp2* KO mice obtained at physiological temperature (**Fig. S1D**), further substantiating the evidence of severity and early onset of the DKO phenotype.

The “non-inactivating” nature of the IHC Ca_v1.3 channels was partially restored in the *Cabp1/2*-DKOs by transgenic expression of WT CaBP2 (Student’s *t*-test: *p* < 0.0001; **Fig. 1B–C**; see also **Fig. S1**). The lack of complete recovery suggests either an inadequate expression of transgenic material or irreversible adaptive changes in response to the lack of major CaBPs that may occur prior to gene delivery. Similarly, Ca²⁺ currents were greatly reduced upon application of short train stimuli and rescued efficiently by the introduction of the transgenic CaBP2 (**Fig. 1D**). However, it has to be noted that the cells in which train stimuli were applied showed a considerably lower calcium current inactivation upon a 500-ms step depolarization than the average (0.36 (*n*=5) vs 0.47 (*n*=21)). Short stimuli at different potentials revealed maximal Ca²⁺-current amplitudes in the IHCs of DKO mice that were comparable to controls (**Fig. 1E**). We observed a slight shift in the voltage-dependence of Ca_v1.3 activation towards more hyperpolarized values (**Fig. 1F**; potential of half-activation – *V*^{0.5} of -33.6 ± 0.5 vs -30.9 ± 0.5 mV in *Cabp1/2* DKO and WT IHCs, respectively; *p* < 0.0001, Student’s *t*-test), and an increased slope of the Boltzmann fit function (7.6 ± 0.1 vs 7.1 ± 0.1, *p* < 0.0001, Student’s *t*-test). This finding is somewhat surprising as the opposite would be expected based on data from heterologous expression systems^{4,12,21}. At present we can not exclude potential changes in the relative amounts of different Ca_v1.3 splice variants, auxiliary subunits or other interaction partners that may in the absence of CaBPs lead to these changes in the IHCs.

To further investigate the nature of the enhanced Ca_v1.3 inactivation, we recorded Ca_v1.3 currents in the presence of 2 mM Ca²⁺ or Ba²⁺ in the extracellular solution (**Fig. 2**), and analyzed the two types of inactivation as described previously²². The lack of CaBP1 and 2 significantly enhanced both, VDI and CDI of IHC Ca_v1.3 channels, which were approx. two- and six-times larger than those in WT IHCs (**Fig. 2B**; VDI: 0.30 ± 0.02 vs 0.13 ± 0.02 and CDI: 0.65 ± 0.01 vs 0.11 ± 0.02 in *Cabp1/2* DKO and WT IHCs, respectively; *p* < 0.0001, Student’s *t*-test).

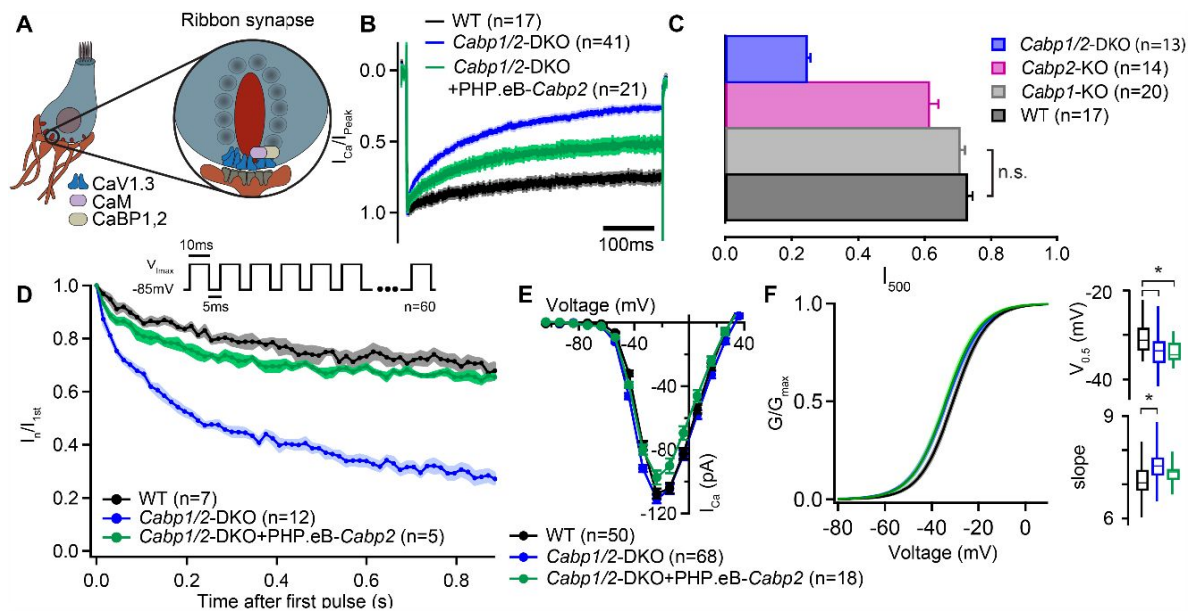


Figure 1.

Strong inactivation of IHC $\text{Ca}_v1.3$ channels upon deletion of *Cabp1* and *Cabp2*.

(A) Schematic of the IHC and one of its several ribbon synapses. CaBPs and CaM modulate voltage-gated calcium channels at the IHC ribbon synapses to shape the presynaptic Ca^{2+} signal. (B) Peak normalized Ca^{2+} -currents after a 500-ms depolarization step to the maximum current potential in the apical IHCs of WT and *Cabp1/2*-DKO animals. Note a pronounced inactivation of the Ca^{2+} current in the absence of CaBP1 and 2 and partial recovery upon *Cabp2* gene replacement. (C) The fraction of the remaining Ca^{2+} -current after a 500-ms depolarization step is depicted for different mouse models acquired at comparable conditions (room temperature, 2 mM extracellular $[\text{Ca}^{2+}]$). The histogram combines the results of this study (*Cabp1/2*-DKO) with previously published data on CaBP1¹¹ and CaBP2¹². The WT data combines pooled data from the current study and controls as obtained previously¹¹. Note that the recordings from *Cabp2*-KO animals were acquired at the lower apex to mid-cochlear tonotopic positions as the phenotype in the apex (low-frequency positions investigated otherwise) is mild¹². (D) Ca^{2+} -currents upon a train of 10-ms long depolarization steps normalized to the amplitude of the first pulse. (E) Ca^{2+} -current-voltage relationships show slightly reduced amplitudes in the IHCs of the *Cabp1/2*-DKO animals transduced with the PHP.eB-*Cabp2*. (F) Activation curves were calculated with Boltzmann fits. Note a significant difference in the half-voltage activation ($V_{0.5}$) and slope in the CaBP1/2-deficient IHCs and the persistence of the voltage shift upon re-expression of CaBP2 (inserts). See main text for the *p* values. Data in B and D-F was acquired in 1.3 mM, data in C in 2 mM $[\text{Ca}^{2+}]$.

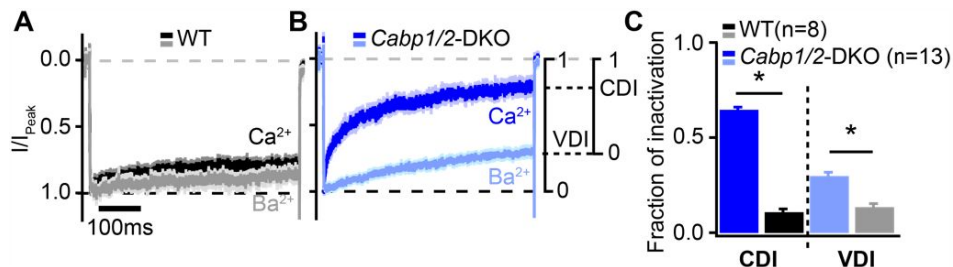


Figure 2.

Voltage- and Ca^{2+} -dependent inactivation are enhanced after deletion of *Cabp1* and *Cabp2*.

(A-B) Peak normalized Ca^{2+} - and Ba^{2+} currents in WT and CaBP1/2-deficient IHCs recorded in the extracellular solution containing 2 mM of the respective divalent cation. The two types of inactivation were calculated as depicted. (C) VDI and CDI were both significantly enhanced in the IHCs of *Cabp1/2*-DKO animals (asterisks; Student's *t* test, $p < 0.0001$).

Impaired synaptic transmission in the IHCs lacking CaBP1 and 2

Since Ca^{2+} influx through $\text{Ca}_v1.3$ channels triggers IHC neurotransmission we next investigated how the prominent inactivation affects exocytosis in the CaBP1/2-deficient IHCs. Exocytosis was probed by recordings of membrane capacitance (**Fig. 3**). In response to short stimuli [primarily recruiting the readily-releasable pool (RRP) of vesicles], increases in capacitance were comparable, while long stimuli (reporting sustained exocytosis) evoked significantly smaller responses in the CaBP1/2-deficient IHCs (**Fig. 3A-B**). These results are in line with the observation of comparable peak Ca^{2+} -current amplitudes but reduced calcium charge transfer (Q_{Ca}) for stimulus durations above 50 ms. However, calculating exocytosis efficiency revealed that the reduction in sustained exocytosis cannot completely be explained by the diminished Ca^{2+} influx (**Fig. 3C**). This suggests an additional impairment in exo- or endocytic processes, the underlying mechanism of which is currently unknown. Transgenic expression of CaBP2 largely restored the efficiency of exocytosis, indicating that the impairment as observed in the IHCs of DKO is related to the absence of the CaBPs. Such exocytosis defect was not observed in *Cabp2*-KO, suggesting that when IHCs are held at hyperpolarized potential at RT, CaBP1 provides sufficient compensation to obscure the synaptic phenotype^{12,20}.

Next, we have reasoned that holding a cell at a relatively hyperpolarized potential of -85 mV as typically utilized in patch-clamp recordings may not reveal the synaptic phenotype in its severity. We thus performed additional sets of experiments (**Fig. 3D-E**), in which the cells were held at either -65 or -55 mV, closer to the suggested resting potential of IHCs²³⁻²⁵. Raising the holding voltage to -55 mV resulted in a significant reduction in the maximal Ca^{2+} currents, which was more pronounced in the CaBP1/2-deficient IHCs as compared to controls ($60 \pm 4\%$ vs $25 \pm 4\%$ reduction in DKO and WT IHCs, respectively; $p < 0.005$, Student's *t*-test). Consequently, further discrepancy between the Q_{Ca} in IHCs of WT and DKO animals was observed ($p < 0.001$), which in DKO animals resulted in a complete loss of sustained exocytosis (**Fig. 3D**; $p < 0.005$). The latter was well restored upon AAV-mediated transgenic expression of CaBP2 ($p < 0.05$), which also reduced the amount of calcium channel inactivation ($38 \pm 8\%$ reduction in the maximal Ca^{2+} current as well as a partial recovery of Q_{Ca} as compared to *Cabp1/2*-DKO IHCs; $p < 0.05$). Rapid voltage oscillations during capacitance recordings may additionally activate calcium channels prior to the depolarizing step. We thus reduced the sine wave amplitude, which resulted in an intermediate phenotype, indicating pronounced activity-dependent deterioration of synaptic function in the absence of CaBPs (**Fig. 3E**). We propose that *in vivo* in the absence of CaBP1 and 2 a large fraction of IHC $\text{Ca}_v1.3$ channels are in a state of prolonged inactivation. This does not support sufficient RRP exocytosis (**Fig. 3D**) to sustain sensory coding during ongoing stimulation of IHCs.

Lastly, we asked whether the deletion of CaBP1 and 2 affects the speed of recovery from $\text{Ca}_v1.3$ inactivation (**Fig. 4**). We applied an initial 5-ms long pulse followed by a conditioning stimulus and a series of 5-ms long pulses to test the kinetics of recovery of the Ca^{2+} -current amplitude. Upon 500-ms depolarization step (**Fig. 4B**), the speed of recovery from inactivation was fast both in DKO as well as WT IHCs where a much smaller fraction of channels underwent inactivation (time constants of 0.62 and 0.75 s in DKO and WT IHCs, respectively; **Table 1**, left). As expected, transgenic expression of CaBP2 reduced the initial inactivation (Ca^{2+} -current amplitude 10 ms after the conditioning pulse, expressed as the fraction of initial current $-I^{10\text{ms_post}}/I^{\text{pre}}$, WT: 0.73 ± 0.01 ; DKO: 0.36 ± 0.03 ; DKO+PHP.eB-*Cabp2*: 0.64 ± 0.02 ; $p < 0.001$ for DKO vs DKO+PHP.eB-*Cabp2*). However, the recovery from the step-evoked inactivation was slowest in the *Cabp2*-injected DKO IHCs. We hypothesize that this may be related to the speed at which the intracellular $[\text{Ca}^{2+}]$ after depolarization is restored to the basal level allowing $\text{Ca}_v1.3$ recovery from CDI (see discussion). A 2-min long sine wave (± 15 mV) from -55 mV to mimic a milder but prolonged hair cell stimulation (**Fig. 4C**) elicited smaller inactivation of $\text{Ca}_v1.3$ channels as compared to a 500-ms depolarization step ($I^{10\text{ms_post}}/I^{\text{pre}}$ of 0.73 ± 0.01 in WT; 0.58 ± 0.03 in DKO; 0.68 ± 0.03 in DKO+PHP.eB-*Cabp2*; $p < 0.05$ for DKO vs DKO+PHP.eB-*Cabp2*). In these conditions, however, the recovery from inactivation was slowest in the DKO IHCs (**Table 1**, right). This suggests that in

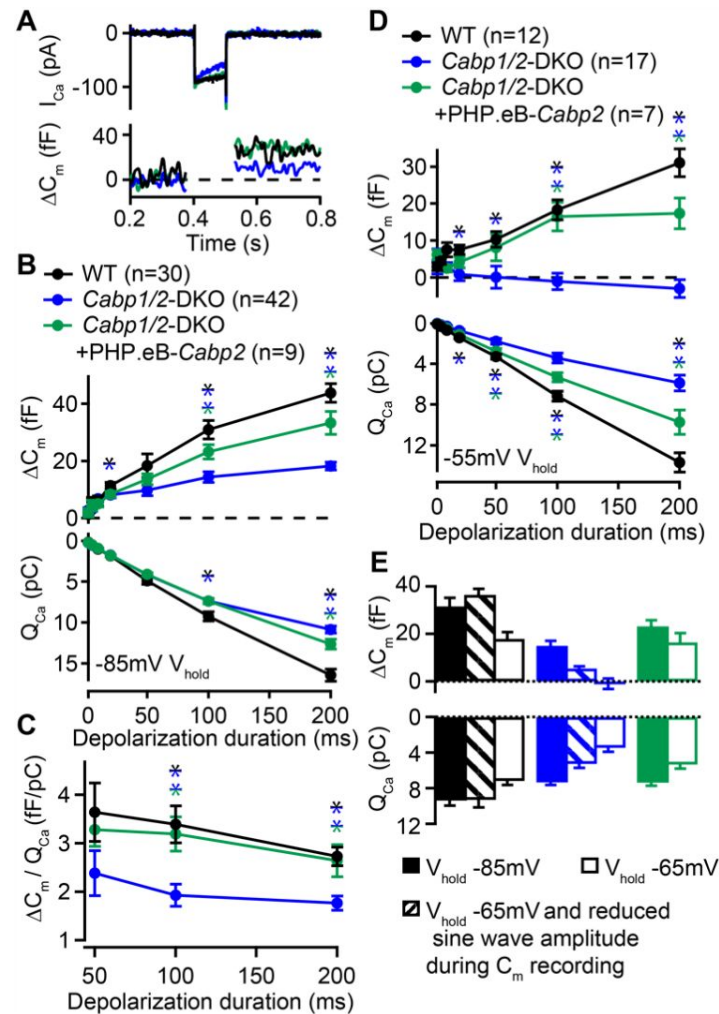


Figure 3.

Reduced sustained component of exocytosis and calcium charge transfer (Q_{Ca}) can be partially rescued by *Cabp2*-transgene-expression.

(A) Representative Ca^{2+} -current traces and corresponding membrane capacitance changes upon 100-ms long depolarization steps to the peak Ca^{2+} -current potential. **(B)** Capacitance increments with the corresponding Q_{Ca} were probed by different depolarization durations from a holding potential of -85 mV (Student's t test or Wilcoxon rank-sum test). *CaBP1/2*-deficient IHCs showed significant impairment of the sustained exocytosis ($p < 0.05$ for 20-ms, and < 0.001 for 100- and 200-ms step, respectively) and reduced cumulative Ca^{2+} influx ($p < 0.005$ for the longest two pulses) as compared to WT controls (blue-black asterisks). AAV-mediated delivery of *Cabp2* improved IHC synaptic function as compared to non-injected *Cabp1/2*-DKO controls ($p < 0.05$ for DC^{m} upon 100- and 200-ms step, and Q_{Ca} upon 200-ms step; green-blue asterisks). **(C)** Note a reduced efficiency of Ca^{2+} -dependent exocytosis ($p < 0.001$ for the two longest test pulses; blue-black asterisks), which can be efficiently rescued by intracochlear delivery of PHP.eB-*Cabp2* ($p < 0.05$ and 0.005 for 100- and 200-ms pulse, respectively; green-blue asterisks). **(D)** Capacitance measurements at a holding potential of -55 mV reveal an aggravation of the phenotype by additional activation of calcium channels between the test pulses ($p < 0.05$ for 10- and 50ms, and $p < 0.005$ for 20-, 100- and 200-ms pulse) and a further discrepancy between the Q_{Ca} of WT and *CaBP1/2*-deficient IHCs (2-10 ms: $p < 0.05$; 20-200 ms: $p < 0.001$). Also in these recording conditions, the IHCs from *Cabp2*-injected *Cabp1/2*-DKO animals showed increased Q_{Ca} (50-200 ms: $p < 0.005$) and exocytosis (100 ms: $p < 0.005$; 200 ms: $p < 0.001$) as compared to non-injected controls. **(E)** Capacitance increments and the corresponding Q_{Ca} upon 100-ms depolarization steps as recorded in different conditions. Note worsening of the phenotype with increasing IHC activation between the test pulses.

the absence of CaBP1 and 2 even mild but prolonged stimulation of IHCs *in vivo* may drive a significant fraction of Ca_v1.3 channels into “deep-state” of inactivation with very slow recovery, as was described for Na_v, K_v and Ca_v channels [26](#)[32](#)–[29](#)[33](#), not allowing efficient sound encoding.

Altered synaptic morphology in *Cabp1/2*-DKO mice

Immunolabeling of the organs of Corti indicated normal presynaptic Ca_v1.3 clustering with no apparent change in size or shape of the clusters (Fig. S2A), and comparable abundance of ribbon synapses in the apical IHCs of WT and DKO animals (12.9 ± 0.2 vs 12.8 ± 0.8 ; respectively) at 3 weeks of age (Fig. S2B–C’). However, the intensities of pre-(RibeyeA/CtBP2) and postsynaptic (Homer1) immunofluorescent puncta were increased in the CaBP1/2-deficient IHCs as compared to age-matched controls, processed in parallel (Fig. S2C’). A similar alteration in the synaptic size was observed in some other synaptic mutants as well as upon noise trauma [30](#)[32](#),[31](#)[33](#), but not in *Cabp2* single KOs [12](#)[33](#), which show significantly less impaired presynaptic Ca²⁺ signalling. This data is in line with compensatory postsynaptic changes in conditions of no or very little postsynaptic activity, possibly governed through adaptive changes in the postsynaptic elements.

As earlier studies of Ca_v1.3 channelopathies demonstrated a lag or stop in IHC development [30](#)[32](#),[32](#)[33](#)[33](#) we next examined the expression of small-(SK) and large-conductance Ca²⁺-activated potassium channels (BK) in the IHCs at different ages. As expected for normal maturation, the SK2 puncta were mostly absent from DKO IHCs by 3 weeks of age and the BK puncta underwent a typical transition from a spotty appearance throughout the cell membrane to the patches at the IHC necks (Fig. S2D–G). However, by 3 weeks and more obviously later on, the BK channel clusters appeared smaller in the DKO IHCs as compared to controls, which could be partially reversed by virally-mediated expression of CaBP2 (Fig. S2F–G). We hypothesize that CaBPs may directly (e.g. as transcription factors) regulate BK expression, a notion that needs to be tested in the future. Alternatively, the effects may be more indirect, as they are common in mutants with impaired intracellular Ca²⁺ signaling [30](#)[32](#),[32](#)[33](#)[33](#).

Severe hearing impairment in mice deficient in CaBP1 and CaBP2

Hearing was assessed by measuring auditory brainstem responses (ABRs; Fig. 5[32](#), Fig. S3). Already in young (3–4-week-old) *Cabp1/2*-DKO animals, ABR thresholds were highly elevated, with tone bursts up to a maximum intensity of 90 dB barely eliciting any ABR response; click responses were better preserved but greatly reduced in amplitude (Fig. S3A–B). At 7–13 weeks of age, further worsening of the click thresholds was observed. Upon 80-dB click stimulation no sizable ABR wave I was detected in average traces of DKO animals at either age tested (Fig. S3B), suggesting reduced and/or desynchronized spiking of spiral ganglion neurons (SGNs). Like in other mouse mutants with defects in the IHC synapse [34](#)[35](#)[35](#), the later ABR peaks were better preserved, indicating partially preserved activation of neurons of the auditory brainstem. To assess cochlear amplification by the outer hair cells (OHCs), we recorded distortion product otoacoustic emissions (DPOAEs). While OAEs were so far only detectable in some of the younger DFNB93 patients [12](#)[36](#)[36](#), mouse work corroborates the notion of at least initially intact cochlear amplification in the absence of CaBP2 [11](#)[12](#)[12](#)[20](#)[37](#). Our results suggest that cochlear amplification is largely preserved also in DKO animals (Fig. S3C–F). In about half of the DKO animals in both age groups, the DPOAEs were normal. In the other half, they were reduced in amplitude but still detectable (Fig. S3E–F), which may suggest partially hampered cochlear amplification in a subset of DKO animals.

Transgenic expression of CaBP2 led to significant improvement of the IHC Ca_v1.3 function and synaptic exocytosis (Figs. 1[32](#)–4[38](#)), we thus next asked, how this may influence the hearing of the animals. As previously observed [20](#)[38](#), unilateral injection of the PHP.eB-*Cabp2*-eGFP resulted in strong eGFP immunofluorescence signal in the vast majority of hair cells and elsewhere (Fig. S4A–

Test group	500-ms depolarization					2-min sine wave				
	A ₁ (%)	τ ₁ (s)	A ₂ (%)	τ ₂ (s)	y ₀ (%)	A ₁ (%)	τ ₁ (s)	A ₂ (%)	τ ₂ (s)	y ₀ (%)
DKO	65	0.62	-	-	99	19	1.36	21	19.41	98
WT	28	0.75	-	-	100	24	3.83	-	-	97
DKO+PHP.eB- <i>Cabp2</i>	22	0.47	14	3.87	99	13	1.88	18	11.56	99

Table 1.

Ca²⁺-current recovery fit parameters.

The data was fitted with the single or the double exponential with the following equations: $y(t) = y_0 - A_1 \times \exp(-(t/\tau_1))$ and $y(t) = y_0 - A_1 \times \exp(-(t/\tau_1)) - A_2 \times \exp(-(t/\tau_2))$, where $y(t)$ represents the peak current amplitude at the time t after the end of conditioning pulse (either 500-ms depolarization or 2-min sine wave) given in % of the initial pre-conditioning peak current amplitude ($100 \times I_{\text{post}}/I_{\text{pre}}$). While current recovery upon 500-ms depolarization in the IHCs of WT and DKO animals could be well fitted by a single exponential, the data obtained in the injected DKO animals required a double exponential fitting. Upon 2-min sine wave, the current recovery in DKO IHCs (both, injected and non-injected group) also required fitting with a double exponential function.

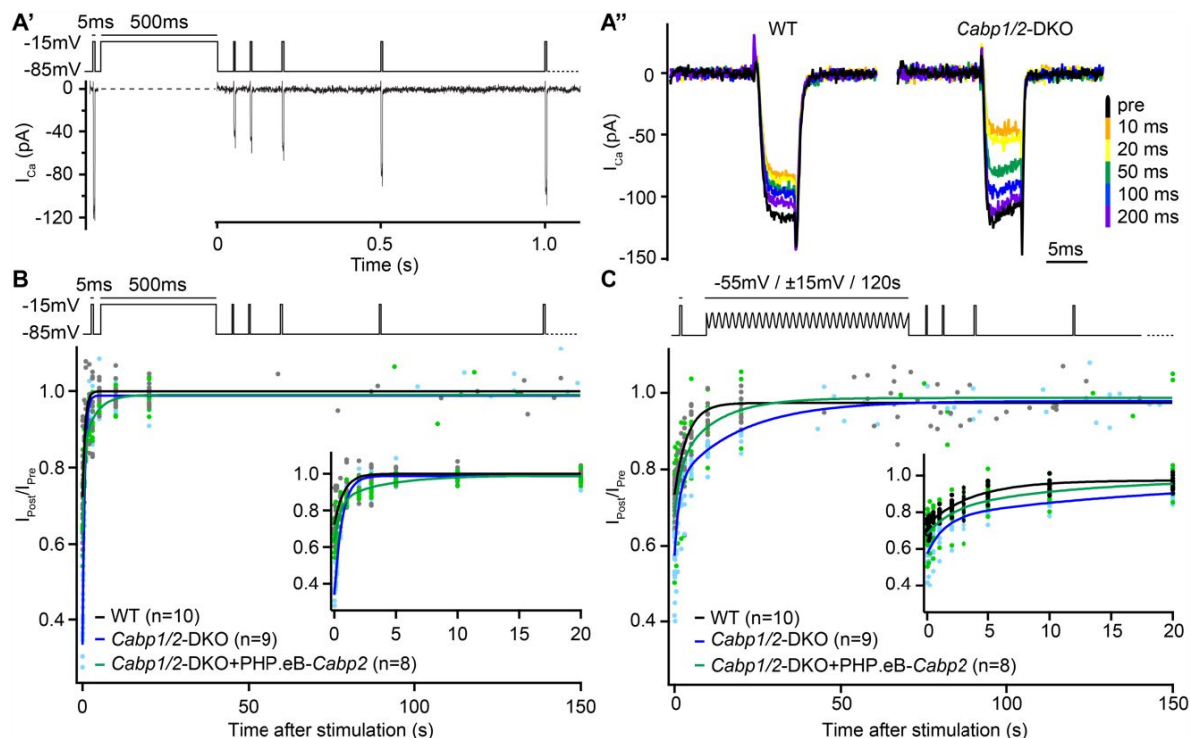


Figure 4.

The speed of recovery of Ca^{2+} currents is affected in *CaBP1/2*-deficient IHCs.

(A) Ca^{2+} currents were probed by 5-ms long depolarization steps to the maximum current potential. An initial test pulse was followed by a conditioning stimulus (a 500-ms depolarization step or a 120-s long sine wave). Subsequently, Ca^{2+} currents were tested at various time points to assess the recovery of Ca^{2+} -current amplitudes. (A') Example Ca^{2+} -current recording using a 500-ms long depolarization step as a conditioning stimulus. (A'') Representative Ca^{2+} -current traces in WT and *CaBP1/2*-deficient IHCs at different time points after a 500-ms long conditioning stimulus. (B-C) Ca^{2+} -current amplitudes normalized to the amplitude of the pre-conditioning test pulse up to 150 s after a 500-ms depolarization step (B) or a 2-min long sine wave stimulus (C). The inserts show the initial 20 s after the conditioning stimulus. The data were fitted by a single or double exponential function (fit parameters in the [Table 1](#)). Note stronger inactivation in the *Cabp1/2*-DKO IHCs and delayed recovery of Ca^{2+} currents after the presentation of a long sine wave and a partial recovery upon viral gene transfer of *Cabp2*. Recovery upon a 500-ms pulse was much faster and slowest in the injected animals.

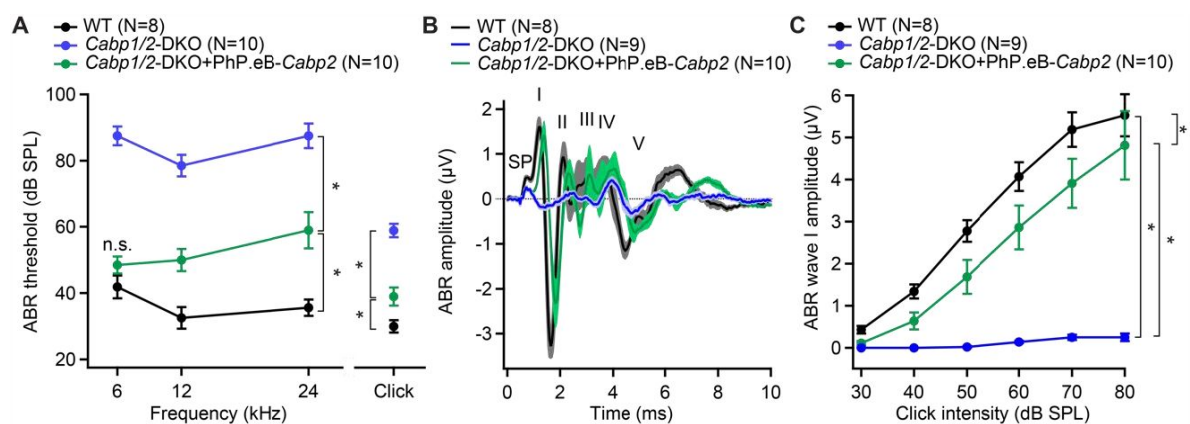


Figure 5.

AAV-mediated transgene-expression of *Cabp2* ameliorates hearing loss and recovers ABR amplitude responses in *Cabp1/2*-DKO mice.

(A) Significant improvement of ABR thresholds in the injected ears of 3-4-week-old *Cabp1/2*-DKO animals for all tone burst frequencies tested as well as 20-Hz-click stimulus (see Table S1 for the p values). (B) Average click-evoked responses to an 80-dB 20-Hz-click stimulus. (C) A strong recovery of the amplitude of the ABR wave I (Tukey's multicomparison test; $p \leq 0.0001$ for DKO vs WT or vs injected DKO) upon *Cabp2*-transgene expression.

B). Transgenic expression of CaBP2 in DKO animals led to significantly improved hearing with lower ABR thresholds, increased click-evoked ABR amplitudes, and shortened wave I latency (**Fig. 5C**, Fig. S4C-D).

Finally, we studied sound encoding by inspecting sound-induced spiking responses of SGNs (**Fig. 6C**). In WT mice, we recorded responses in 39 neurons, which were classified as SGNs based on their location, response patterns, spike waveforms and recording stability. Their spiking characteristics resembled typical WT data ^{12,37,40}. In response to 50- or 500-ms tone burst at the characteristic frequency 30 dB above threshold (i.e. where the sigmoidal rate-intensity functions have saturated spike rates) and at a stimulus repetition rate of 5 Hz, they showed the SGN-typical very high onset firing rates, gradually adapting to reach steady-state rates between 100 and 400 Hz (**Fig. 6A-B**). In contrast, only one out of the 20 neurons we recorded and analyzed from DKO animals showed a normal response in our automated tuning curve algorithm with a threshold of 35 dB at 7.8 kHz, while all others displayed thresholds above 80 dB and extremely low spike rates (**Fig. 6A-B**). Since in the majority of neurons frequency tuning was very broad, we used subjectively chosen best frequency tone burst or white noise burst stimulation of at least 80 dB for subsequent characterization. 16 out of 19 DKO units responded with a relatively high onset rate but strong adaptation to a minimal sustained spiking (**Fig. 6A-B**), which prevented unit classification based on the response patterns. Judging from other criteria described above, we assume that approximately 13 of these represent SGNs and 3 represent bushy cells or multipolar cells of the anteroventral cochlear nucleus. Three additional units (two resembling a pauser/buildup- and one displaying a clear chopper neuron-type response) were excluded from further analysis. Overall, in DKO neurons, peak and adapted rates were greatly reduced as compared to WT controls (**Fig. 6C**), which came along with an increased latency (12.6 ± 2.8 ms vs WT 4.8 ± 0.2 ms; $p < 0.001$, Wilcoxon rank-sum test) and increased jitter (18.2 ± 5.0 vs WT 1.5 ± 0.4 ; $p < 0.00001$, Wilcoxon rank-sum test) of the first spike that followed the spike onset.

DKO responses greatly improved when we lowered the stimulus repetition rate to 0.5 Hz, and plateaued at 0.2 Hz (**Fig. 6C-D**). We hypothesize that in DKO SGNs, recovery of $\text{Ca}_v1.3$ channels after a silent interval in the range of 1-2 seconds is the limiting factor for determining the maximal SGN spiking rate at the sound onset. This may be further contributed by potentially slowed re-supply of synaptic vesicles, as suggested by reduced efficiency of sustained exocytosis in the CaBP1/2-deficient IHCs (see **Fig. 3**). In contrast, in WT SGNs, the maximal spike rates are mostly recovered after a silent interval of 150 ms, which may be mostly limited by vesicle replenishment at the IHC ribbon synapse ^{35,41}. During a 5-Hz sound stimulation, the strength of adaptation was greatly enhanced (peak to adapted ratio 12.5 ± 3.0 vs WT 3.4 ± 0.1 ; $p < 0.00001$, Wilcoxon rank-sum test) whereas the time constant of adaptation was unchanged (7.9 ± 2.0 ms in DKO compared to 7.0 ± 0.4 ms in WT ($p = 0.5$, Wilcoxon rank-sum test). However, for the longer 500-ms tone bursts used in the 0.5 Hz stimulation dataset, it becomes obvious that despite the strong adaptation, DKO neurons are still able to sustain spiking at a low rate throughout the stimulation (**Fig. 6B**). DKOs also showed reduced spontaneous spiking activity (**Fig. 6E**), which supports the hypothesis that already in the absence of sound stimulation over tens of seconds a considerable fraction of calcium channels may be in steady-state inactivation.

The reduction in spike rates and their strong dependence on the strength of stimulation in the preceding seconds introduces experimental bias to our data. In the units of the DKO animals, it was not feasible to faithfully assess sound thresholds or dynamic ranges. Furthermore, the overall experimental yield was generally much lower in DKO animals. On average we hit 0.3 sound-responsive neurons per hour of recording, compared to 2.5 in WT. Very likely many DKO neurons failed to generate spontaneous and evoked action potentials, thus we were not able to identify them. Based on the DPOAE data we assume that cochlear mechanics and active amplification and thus the IHC receptor potential thresholds are mostly normal in the DKOs. We cannot exclude that

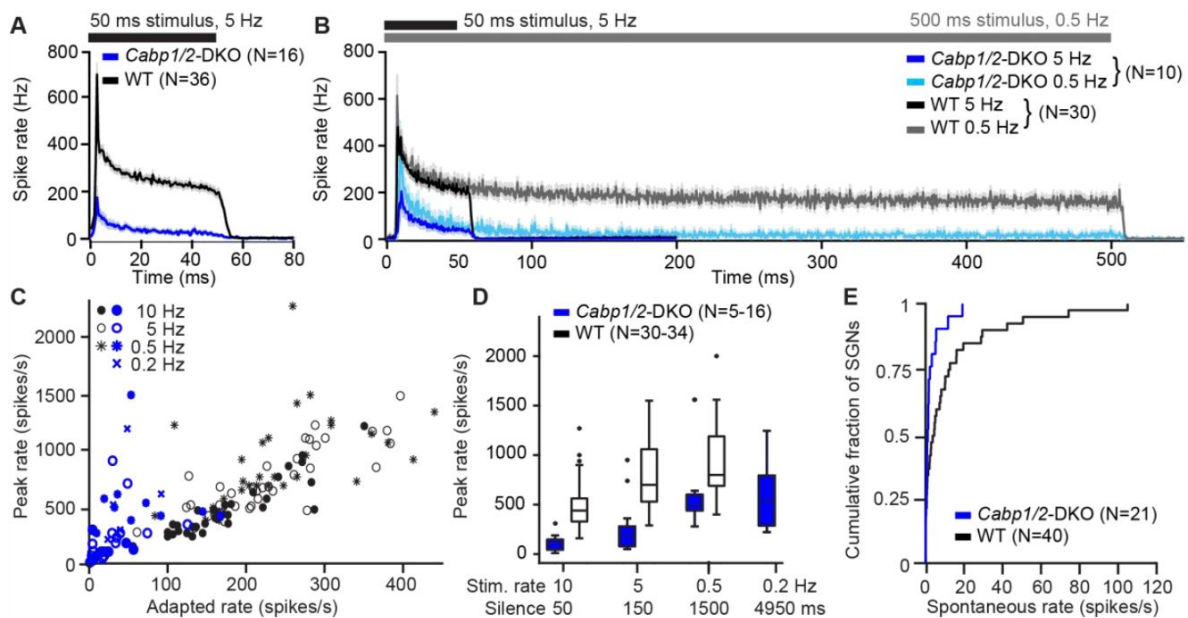


Figure 6. Nearly blocked sound encoding in *Cabp1/2*-DKO mice. Single neuron responses to

Figure 6.

Nearly blocked sound encoding in *Cabp1/2*-DKO mice.

Single neuron responses to sound stimulation *in vivo* in the region of the auditory nerve of *Cabp1/2*-DKO mice were extremely scarce. **(A)** WT poststimulus time histograms of SGNs to 50-ms suprathreshold tone or noise burst stimuli presented at a rate of 5 Hz showed a very high sound onset rate gradually adapting to a sustained rate persisting throughout the stimulus duration. Spike rates were much lower in the mutants. **(B)** Applying a lower stimulus rate of 0.5 Hz partially restored sound onset responses in *Cabp1/2*-DKOs, highlighting the enhanced strength of adaptation. WT and mutant neurons sustained their adapted spiking response throughout the longer 500 ms stimulus used in these recordings. For direct comparison, the average 5-Hz poststimulus time histograms from the same neurons are shown. **(C)** Correlations of peak and adapted rates for different stimulus rates. **(D)** Peak rates for different stimulus rates / interstimulus intervals in WT SGNs strongly increase between 10 and 5 Hz, while in DKOs a longer period of silence is required to observe such difference. **(E)** *Cabp1/2*-DKO neurons had lower spontaneous spike rates than WT SGNs.

IHC receptor potentials are decreased in amplitude, which however should rather reduce SGN adaptation. We thus interpret the observed increase in adaptation and delay of recovery of SGN spiking to be the effects of changed synaptic $\text{Ca}_v1.3$ channel kinetics as described above.

In summary, sound responses of DKO SGNs and cochlear nucleus neurons showed a drastic reduction in spontaneous and evoked action potential rates, which was associated with increased sound thresholds and reduced temporal precision of action potential generation. Together, these factors explain well the poor ABR thresholds (Fig. 5 [↗](#), Fig. S4A).

Discussion

In this study, we examined the effects of genetic disruption of the two major CaBPs expressed in the cochlea. In the IHCs of CaBP1/2-deficient animals, we found largely inactivated Ca^{2+} currents with prolonged recovery from inactivation upon mild oscillatory stimulation, which did not support sufficient neurotransmission for proper sound encoding. The extent of channel inactivation and impairment of IHC synaptic exocytosis aggravated with the extent of IHC stimulation. Similarly, SGN action potential rates *in vivo* declined at increased sound stimulation rates and displayed very strong adaptation upon stimulation, going along with impaired sustained IHC exocytosis. The impairment was significantly stronger as compared to the phenotype in the absence of CaBP2 alone. This suggests that under *in vivo* conditions the roles of CaBP1 and 2 are partially overlapping.

Function and expression profile of CaBPs in the cochlea

The murine cochlea shows strong expression of CaBP1 and 2, while the levels of CaBP4 and 5 seem negligible ^{19 [↗](#), 42 [↗](#), 43 [↗](#)}. Among the three putative splice isoforms of CaBP2 differing in the N-terminal aminoacid sequence, the alternative long isoform, termed CaBP2-alt, may be the most abundant splice variant in the IHCs, as suggested by *in situ* hybridization ^{19 [↗](#)}, while SGNs seem to express none. So far our gene-rescue experiments were conducted with the conventional L-isoform only, which however shows strong inhibition of $\text{Ca}_v1.3$ inactivation in the HEK cells as well as IHCs ^{12 [↗](#), 20 [↗](#)}. Alternative splicing also generates three CaBP1 variants ^{19 [↗](#)}. The alternative long variant caldendrin is primarily expressed in the SGNs or the surrounding glia cells, while weak expression of CaBP1-S/-L (as judged by the mRNA production) was detected in hair cells ^{19 [↗](#)}. Our DKO model lacks all splice isoforms of both CaBPs. Since *Cabp1* KO animals show normal hearing several weeks after birth, and CaBP2 is absent from SGNs, we interpret the hearing impairment of the *Cabp1/2* DKO animals to be primarily due to defects of hair cell function and judge contribution by SGN dysfunction as rather unlikely, at least in the early postnatal weeks.

The ABR thresholds of *Cabp1* KO animals are normal at 4 weeks of age, but deteriorate with age, starting with high-frequency hearing loss at 9 weeks. Conversely, the suprathreshold ABRs to tone burst stimulation show increased amplitudes and reduced latencies during the first two months after birth. This suggested enhanced SGN activity at first, followed by slowly progressing functional decline, possibly due to overexcitability-related toxicity ^{11 [↗](#)}. Due to early strong impairment of hair cell synaptic function, the latter effects are not expected to occur in the DKO animals. However, ABR thresholds for clicks were further increased and an additional reduction in the summing potential was observed in the DKO animals at two as compared to one month of age despite relatively well-preserved DPOAEs (Fig. S4). It thus remains to be clarified whether the absence of CaBP1 and 2 may affect (IHC) mechanotransduction at more mature stage. Unfortunately, this is difficult to assess due to the increased fragility of hair bundles at mature age. Intriguingly, CaBPs share similarities with calcium and integrin-binding protein 2 (CIB2), associated with hair bundle defects ^{44 [↗](#)–46 [↗](#)}. When we overexpressed CaBP2 in the DKO animals, interestingly, suprathreshold responses of ABR wave I were very large (Fig. 5 [↗](#)) and the

summated wave I-IV responses were not distinguishable from WT responses. However, tone burst and click thresholds did not completely recover to WT levels. The difference in the extent of rescue of threshold and suprathreshold responses may be due to possibly distinct requirements for CaBPs at synapses providing inputs to low- or high-threshold SGN fibers. Apart from a small shift in voltage activation and abundance of the channels⁴⁷, not much is known about the possible differences in the Ca_v1.3 properties and interaction partners at distinct ribbons. Neither is it known whether the composition of the channel complex [i.e. proportions of alpha subunit splice variants^{18,48,49} and different beta subunits⁵⁰] differs systematically between ribbon-AZs partnered by different classes of SGN fibers. Investigation of intracellular Ca²⁺ signals from individual AZs may offer some answers in the future. Different amounts of rescue of threshold vs suprathreshold responses could be further due to confounding effects of CaBP1 and 2, e.g. augmented SGN response in *Cabp2*-injected DKO animals still lacking CaBP1 (reminiscent of the phenotype of young *Cabp1* KOs) in combination with “only” partial rescue of CaBP2 function. Possible adverse effects of exogenous expression of CaBP2, e.g. off-target toxicity and uncontrolled overexpression of the transgene, have been discussed previously²⁰, and further experiments need to be performed in the future to achieve (close to) complete rescue of hearing function in these mutants.

Our results reveal that the amount of Ca_v1.3 inactivation increases dramatically when IHCs lack both major CaBPs. This suggests that the sole loss of either CaBP1 or 2, can be largely compensated by the other of the two CaBP members. Yet, the loss of CaBP2 alone still results in progressive hearing impairment in mice and DFNB93 in humans. The inability of CaBP1 to fully replace CaBP2 in the CaBP2-deficient IHCs may stem from either too low concentration of CaBP1 (and other CaBPs), or from distinct properties of the two proteins (enabling CaBP2 to exert more potent suppression of Ca_v1.3 inactivation, e.g. VDI, or potentially performing additional, yet unidentified cellular function). Interestingly, even the overexpression of CaBP2 did not completely rescue the hearing phenotype of DKO animals. Future experiments involving overexpression of CaBP1 (and possibly CaBP1-2 molecular chimeras) in *Cabp2* SKO and/or *Cabp1/2* DKO animals together with further biophysical interaction studies are required to gain a better understanding of the potentially distinct functions performed by the two major CaBPs in the hair cells.

CaBPs as the regulators of IHC synaptic transmission

Previously, patch-clamp recordings surprisingly revealed normal whole-cell exocytosis in CaBP2-deficient IHCs despite enhanced inactivation of calcium channels¹², and no obvious defects in the CaBP1-deficient IHCs¹¹. In contrast, IHCs deficient in both, CaBP1 and 2, displayed strongly impaired exocytosis. Transgenic overexpression of CaBP2 in IHCs resulted in a partial recovery of the Ca²⁺ influx and an even better recovery of exocytosis. Together, these results corroborate the hypothesis of the role of CaBPs in preventing extensive steady-state inactivation of Ca_v1.3 channels to support sufficient synaptic exocytosis for proper sound encoding¹². The lack of CaBP1 and 2 seems to impair sustained exocytosis and efficiency of exocytosis beyond the levels that could be attributed to reduced Q_{Ca}. This became particularly apparent when the IHCs were challenged by higher holding potentials in combination with large sine wave amplitudes during capacitance measurements, which we assume to partially deplete the pool of RRP vesicles. At present, it is not clear how CaBPs may directly affect the exocytosis, e.g. by influencing the coupling of the Ca_v1.3 channels and synaptic vesicles or potentially affecting the expression levels of other synaptic proteins that mediate synaptic vesicle replenishment. In this respect, it is worth noting that CaBP1 and 2 have been reported to modulate CaMKII activity², which in turn may through phosphorylation modify the Ca²⁺ affinity of otoferlin C2 domains⁵¹. In the future, it is thus essential to investigate if CaBPs regulate the function of otoferlin and thus synaptic processes. Furthermore, other members of the CaBP/CaM family have been suggested to interact with synaptic proteins and/or be involved in the regulation of synaptic vesicle cycle: CaM in endocytosis⁵², CaBP5 in priming via interaction with Munc18-1⁵³, CaBP1 via inhibition of IP₃ receptors^{54,55}. An important challenge for future studies is to identify further CaBP(2)-interaction partners and possible functions beyond modulation of calcium channels.

The recovery of $\text{Ca}_v1.3$ channels from inactivation upon prolonged milder stimulation (2 min sine wave) was following a single exponential course in WT IHCs, but additional slower component was detected in the IHCs deficient in CaBP1 and 2. In the IHCs of injected DKO animals, this second component appeared smaller and faster, suggesting that CaBPs speed-up recovery of $\text{Ca}_v1.3$ channels from inactivation. Upon a comparably shorter (500 ms) but stronger depolarization, the recovery of $\text{Ca}_v1.3$ from inactivation was very fast in DKO IHCs, in fact faster than in (WT or) *Cabp2*-injected DKO IHCs. We hypothesize this to be due to faster Ca^{2+} clearance from the DKO IHC synapses as the intracellular $[\text{Ca}^{2+}]$ levels are expected to rise significantly less in CaBP1/2-deficient IHCs due to fast and pronounced $\text{Ca}_v1.3$ inactivation. In contrast, higher $[\text{Ca}^{2+}]$ may buildup at the synapses of WT and “rescued” DKO IHCs; consequently, CDI may be prolonged. Another plausible explanation for the observation is, if we consider that CaBP2 may act as a slow mobile calcium buffer ¹². In the presence of excessive “overexpressed” CaBP2 in injected IHCs of DKO animals, the depolarization-evoked intracellular $[\text{Ca}^{2+}]$ transients may become biexponential as has been described for parvalbumin alpha ⁵⁶, which may prolong the time constant of Ca^{2+} clearance and determine the kinetics of channel recovery from inactivation. Our results of the recovery upon 500-ms depolarization may thus mostly reflect the kinetics of the Ca^{2+} clearance from the cytosol, while the experiments with prolonged sine wave stimulation (when e.g. the function of calcium pumps and other extrusion systems and intracellular $[\text{Ca}^{2+}]$ levels may be at equilibrium), may report the speed of calcium channel recovery from a slow inactivation. Slow inactivation with very slow recovery was described for Na_v , K_v and Ca_v channels ^{26–29}. It has to be noted that the time constant of recovery of $\text{Ca}_v1.2$ from slow VDI was comparably shorter (4 s) as observed in the present study ²⁹. Possibly different conditions (e.g. stimulus parameters) or channel subtype may underlie this difference. Future experiments combining patch-clamp recordings with calcium imaging should be employed to address this question further.

From our *in vitro* recordings at higher holding potential or upon prolonged stimulation with small voltage oscillations we predict that *in vivo* a noteworthy proportion of $\text{Ca}_v1.3$ channels may be effectively inactivated and not available for mediating IHC synaptic transmission. This could explain the low experimental yield, the low “spontaneous” spiking rates of the SGNs and lower sound-evoked spike rates with higher thresholds. Similarly, thresholds may be high for tone bursts in the ABR measurements due to high proportion of inactivated channels upon stimuli at high repetition rate used for these recordings (20-40 Hz repetition rate for click and tone burst stimulation, respectively). We believe that any sound stimulation and potentially even spontaneous activity may effectively silence DKO synapses. As expected, longer interstimulus intervals may allow a larger proportion of channels to recover between stimuli, increasing the onset rates of SGN spiking while decreasing the first spike latency and jitter. Still, high-intensity stimulation had to be used in the DKO animals to evoke an increase in SGN spiking. It is possible that spontaneous activity and/or our routines used to determine the SGN characteristic frequency inactivate a majority of IHC $\text{Ca}_v1.3$ channels thus not allowing reliable assessment of the characteristic frequency. In addition to $\text{Ca}_v1.3$ channel inactivation, other factors may contribute to low hearing sensitivity, such as possible impairment of synaptic vesicle replenishment, or IHC mechanotransduction, and, in some cases, impaired cochlear amplification. A complete lack of BK channels was shown to affect the IHC resting potential, receptor potential amplitude and timing, and consequently the timing and frequency of spiking in the SGNs ⁵⁷. Lower expression of BK channels is not expected to significantly affect hearing thresholds, at least initially ^{58–60}, and thus likely does not contribute to the hearing defect of the *Cabp1/2* DKO animals. However, at advanced ages, the effects described above may combine with the lack of CaBPs to additionally desynchronize already scarce spikes in the auditory nerve fibers.

Clinical implications for the diagnosis and treatment of DFNB93

The SGN sound encoding phenotype in *Cabp2*-KO mice was surprisingly mild compared to the deficits observed in DFNB93 patients e.g. ^{14,12}; SGN thresholds and evoked spike rates were only minimally affected ¹². Given the strong recovery phenotype in the DKO we now consider the possibility that synaptic activity in response to everyday sounds could already inactivate calcium

channels and reduce SGN spiking. In this case, communication in daily life would be more strongly impaired than audiometric data obtained in a soundproof chamber, especially when short stimuli separated by silent intervals are used. In mild cases, a gap detection test may be used to assess auditory fatigue. Based on our data we further predict that patients that have a stronger channel inactivation phenotype might not benefit much from sound amplification by hearing aids. Instead, they should use technologies that improve the signal to noise ratio (e.g. noise cancellation, directional microphones, wireless transmitters). An alternative or additional future treatment approach may be gene therapy. In our hands, re-expression of CaBP2 by genetic treatment partially restored Ca_v1.3 properties, IHC exocytosis and consequently ABR wave amplitudes and thresholds. This successful viral rescue supports the view that a reversible inactivation of IHC Ca_v1.3 currents is the main mechanism of hearing loss in CaBP-related deafness, which can be targeted by gene therapy in the future.

Materials and Methods

Animals

For the generation of the *Cabp1/Cabp2*-double-knockout (*Cabp1/2*-DKO) line, *CaBP2^{LacZ/LacZ}* mice, carrying LacZ trapping cassette in place of the exons 3 and 4 [12](#), were crossbred with *Cabp1* KO mice [61](#) and maintained on a C57/Bl6 background. Mice from 3-13 weeks were used for electrophysiological and systems physiology recordings. All experiments complied with the national animal care guidelines and were approved by the University of Göttingen Board for animal welfare and the animal welfare office of the state of Lower Saxony.

Systems physiology recordings: ABR and DPOAE

Prior to the experiments mice were weighed and anesthetized via intraperitoneal injections of ketamine (125 µg/g) and xylazine (2.5 µg/g). Recordings were obtained in a custom-made soundproof box with a Tucker-Davis (TDT) BioSig System II/BioSig software (Tucker-Davis Technologies, Alachua, FL). Body temperature was maintained using a heating pad (37°C) and ECG was monitored continuously. For ABR recordings, subcutaneous needles were placed at the vertex, the ipsilateral pinna and at the lower back. Using a JBL2402 speaker in free-field configuration, tone bursts (10-ms plateau, 1-ms cos² rise/fall) or clicks of 0.03 ms were ipsilaterally presented at 40 Hz for tone bursts or 20 Hz for clicks. ABR traces were amplified 50000-fold, band-pass filtered between 400 Hz and 4 kHz (NeuroAmp) and averaged from 1300 repetitions sampled at 50 kHz for 20ms. Thresholds were defined in stacked waveforms as the lowest stimulus intensity (dB SPL) at which a reproducible waveform could be visually detected with a 10 dB precision. Wave I amplitudes were measured peak-to-trough. DPOAEs were measured using a custom-written MATLAB (MathWorks) routine. Two primary tones f₁ and f₂ (ratio f₁/f₂ = 1.2, level difference f₂ = f₁ + 10 dB) were presented via two speakers (MF-1, TDT) and a custom-made ear probe containing an MKE-2 microphone (Sennheiser). Stimulus duration was 16 ms and f₂ levels varied from 10 to 70 dB SPL in 10 dB steps. The microphone signal was amplified (DMX 6Fire; Terratec (young animals), or UAC-2; Zoom (older animals)) and digitized (TDT System III). DPOAE amplitude was analyzed at 2×f₂-f₁ using custom-written MATLAB software (MathWorks) with fast Fourier transformation. DPOAE thresholds were determined as responses that exceeded -18 dB SPL in young and -4 dB SPL in older animals.

Auditory nerve fiber recordings

Mice were anesthetized by injection of xylazine (5 mg/kg), urethane (1.32 mg/kg) and buprenorphine (0.1 mg/kg) i.p., tracheotomized, and placed in a stereotactic apparatus. The left occipital bone and the cerebellum were partially removed to visualize the left superior semicircular canal. The latter served as a landmark to navigate a glass microelectrode through the cochlear nucleus, aiming at the location where the auditory nerve leaves the internal auditory

canal and enters the cochlear nucleus. 85 or 90 dB broadband noise bursts (ScanSpeak, Avisoft Bioacoustics) were used as search stimuli to identify neurons that displayed sound-driven spiking activity. They were characterized by spontaneous rate and frequency tuning and the responses to tone bursts at the characteristic frequency, 30 dB above the threshold, as previously described³⁵. In the majority of sound-responsive DKO neurons, thresholds were so high and tuning so broad that neither the automated tuning curve algorithm nor systematic or manual changes in sound frequency elicited frequency-specific responses. In these cases, we either chose a subjective “best frequency” or used white noise burst stimulation at 85 dB.

Patch-clamp recordings

For patch-clamp recordings, the apical coils of 3-4-week-old animals were dissected in HEPES-Hanks solution containing (in mM): 10 HEPES, 5.35 KCl, 141.7 NaCl, 0.5 MgSO₄, 1 MgCl₂, 11.1 D-glucose and 3.42 L-glutamine. pH was adjusted to 7.2. IHCs of the ~6 kHz tonotopical region were investigated using the perforated patch configuration. The pipette solution contained (in mM): 130 Cs-gluconate, 10 TEA-Cl, 10 4-AP, 1 MgCl₂, 10 HEPES, 300 mg/mL amphotericin B, pH 7.2 ~290 mOsm. The bath solution was composed of (in mM): 111 NaCl, 35 TEA-Cl, 2.8 KCl, 1 CsCl, 1 MgCl₂, 10 NaOH-Hepes, 11.3 D-glucose, and 1.3 or 2 CaCl₂ or BaCl₂ (pH 7.2, 300–310 mOsm). When the bath solution contained 2 mM BaCl₂/CaCl₂, NaCl was adjusted to 110 mM. The bath solution was supplemented by the SK2 channel blocker apamin (100 nM; Peptanova, Germany) freshly on the day of the experiment. Recordings were performed at room temperature using an EPC-9 amplifier (Heka-Germany) controlled by Pulse software. Currents were leak-corrected using a p/10 protocol, sampled at 50 kHz and corrected offline for the liquid junction potential (-15 mV). IHCs were patched at a holding potential of -85 mV (in a subset of experiments, -65 and -55 mV) and stimulated with depolarizations to the peak Ca²⁺-current potential. Changes in membrane capacitance were monitored by applying a 2-kHz-sinewave with an amplitude of 35 mV. In a subset of experiments, the amplitude was reduced to 15 mV to diminish possible stimulation of the cells caused by the sinewave. VDI and CDI were calculated as follows: $VDI = 1 - I_{Ba_500}/I_{Ba_initial}$ and $CDI = 1 - (I_{Ca_500}/I_{Ca_initial})/(I_{Ba_500}/I_{Ba_initial})$, where I_{Ba_500} and I_{Ca_500} represent whole cell Ba²⁺ and Ca²⁺ currents 500 ms after beginning of depolarization, and $I_{Ba_initial}$ and $I_{Ca_initial}$ correspond to the initial, peak currents with the respective divalent cations²².

Immunohistochemistry

Apical coils (6-8 kHz tonotopical region) of the organ of Corti were fixed with 4% (vol/vol) formaldehyde in PBS on ice. After incubation for 1 h at room temperature in goat serum dilution buffer [16% (vol/vol) normal goat serum, 450 mM NaCl, 0.6% Triton X-100, 20 mM phosphate buffer (pH 7.4)], the explants were incubated in primary antibodies over night at 4 °C. The following antibodies were used: chicken anti-Homer1 (1:200, Synaptic Systems), rabbit anti-Myosin6 (1:200, Proteus), guinea pig anti-Synapsin1/2 (1:200, Synaptic Systems), rabbit anti-RibeyeA (1:200, Synaptic Systems), guinea pig anti-parvalbumin-α (1:200; Synaptic Systems), rabbit anti-BK (1:100; Alomone), rabbit anti-SK2 (1:200; Alomone), mouse IgG1 anti-CtBP2 (1:200; BD Biosciences), rabbit anti-GluA3 (1:200; Santa Cruz), rabbit-anti-CaV1.3 (1:100; Alomone), chicken anti-GFP (1:200, Abcam). The explants were then incubated with secondary AlexaFluor-labeled antibodies (1:200; Molecular Probes) for 1 h at room temperature, washed 3 times in PBS and mounted. Confocal images were acquired with an Abberior Instruments Expert Line STED microscope using a 1.4 NA 100x oil immersion objective. ImageJ software (NIH) was employed to analyze 2D confocal stacks and generate maximum projections. Volumetric synapse analysis ran on Imaris (Bitplane, Zurich, Switzerland) paired with a custom-written Matlab software⁶².

Molecular biology and AAV vector production

The construct containing the long isoform of mouse CaBP2 (CaBP2-L; NCBI Reference Sequence: NP_038906.2) and eGFP is depicted in Fig. S1A. AAV9-PHP.eB particles were generated using our standard AAV purification procedure previously described in more detail in ref.⁶³. In brief,

triple transfection of HEK-293T cells was performed using pHelper plasmid (TaKaRa/Clontech), the *trans*-plasmid providing viral capsid PHP.eB (a generous gift from Viviana Gradinaru, Addgene plasmid #103005) and the *cis*-plasmid providing hCMV/HBA_wtCaBP2-P2A-eGFP. PHP.eB viral particles were harvested from the medium 72 h after, and from cells and the medium 120 h after transfection. Precipitation was done with 40% polyethylene glycol 8000 (Acros Organics, Germany) in 500 mM NaCl for 2 h at 4°C. Both, precipitate and cells were lysed in high salt buffer (500 mM NaCl, 2 mM MgCl₂, 40 mM Tris-HCl, pH 8,0) and non-viral DNA was degraded using salt-activated nuclease (SAN, ArctiCzymes, USA). The cell lysates were clarified by centrifugation at 2000×g for 10 min and then purified over iodixanol (Optiprep, Axis Shield, Norway) step gradients (15, 25, 40, and 60%) at 350000×g for 2.25 h^{64,65}. Finally, viral particles were concentrated using Amicon filters (EMD, UFC910024) and formulated in sterile PBS supplemented with 0.001% Pluronic F-68 (Gibco, Germany). Virus titer was 8.2×10^{12} genome copies/ml measured by determining the number of DNase I resistant genome copies using qPCR (StepOne, Applied Biosystems) and AAV titration kit (TaKaRa/Clontech). The purity of viruses was routinely checked by silver staining (Pierce, Germany) after gel electrophoresis (NovexTM 4-12% Tris-Glycine, Thermo Fisher Scientific).

Virus injections

Mice were injected at postnatal day 6 using the round window approach as described in earlier studies⁶⁶. In brief, anesthesia was established with isoflurane (5% for induction, 2-3% for maintenance, frequent testing of the absence of hind-limb withdrawal reflex). For analgesia, buprenorphine (0.1 mg/kg, 30 min before surgery) and carprofen (5 mg/kg, during and 1 day after surgery) were applied subcutaneously and Xylocain (10 mg spray) locally. Body temperature was maintained by a custom-built heating blanket. Following a retro-auricular approach, the facial nerve was exposed to determine where to puncture the cartilaginous bulla with the injection pipette and target the scala tympani where virus suspension (1-1.5 µl, corresponding to 0.82 - 1.23×10^{10} AAV particles) was injected. Following injection, the endogenous tissue was relocated and the surgical situs closed by suturing the skin. 2-3 weeks later, the injected mice were used for hearing tests, *in vitro* electrophysiology and immunohistochemistry of the cochleae.

Data Analysis

Data were analyzed using Igor Pro Software (Wavemetrics), GraphPad PRISM (Graphpad Software, Inc.), Matlab (Mathworks) and Imaris (Bitplane). All data are presented as mean ± SEM. For the analysis of half-max Ca²⁺ current (V^{half}) and activation constant (k^{act}), current-voltage traces were transformed into an activation function ($I^{Ca} = g^{max}(V - V^{rev})$), which was subsequently fitted by a Boltzmann function: $1/(1 + \exp((V^{0.5} - V)/k^{act}))$. G^{max} describes the maximum conductance, V the holding potential and V^{rev} the reversal potential. Data were tested for randomness, normality (Jaque-Bera test), equality of variances (F-test) and finally compared for statistical significance by performing Student's *t*-test for normally distributed data of equal variance or else by the Wilcoxon rank sum test. Systems physiology data were tested for significance using ANOVA with Tukey's or Šidak's multicomparisons tests.

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Article and author information

David Oestreicher

Experimental Otolology Group, InnerEarLab, Department of Otolaryngology, University Medical Center Göttingen, 37075 Göttingen, Germany, Auditory Neuroscience Group, Max Planck Institute for Multidisciplinary Sciences, 37075 Göttingen, Germany

Shashank Chepurwar

Auditory Systems Physiology Group, Institute for Auditory Neuroscience, InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany, Collaborative Research Center 889, University of Göttingen, 37075 Göttingen, Germany

Kathrin Kusch

Functional Auditory Genomics, Institute for Auditory Neuroscience, University Medical Center Göttingen, 37075 Göttingen, Germany, Institute for Auditory Neuroscience and InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany

Vladan Rankovic

Institute for Auditory Neuroscience and InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany, Restorative Cochlear Genomics Group, Auditory Neuroscience and Optogenetics Laboratory, German Primate Center, 37075 Göttingen, Germany

Sangyong Jung

Institute for Auditory Neuroscience and InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany

Nicola Strenzke

Auditory Systems Physiology Group, Institute for Auditory Neuroscience, InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany, Collaborative Research Center 889, University of Göttingen, 37075 Göttingen, Germany
ORCID iD: [0000-0003-1673-1046](https://orcid.org/0000-0003-1673-1046)

Tina Pangrsic

Experimental Otolology Group, InnerEarLab, Department of Otolaryngology, University Medical Center Göttingen, 37075 Göttingen, Germany, Auditory Neuroscience Group, Max Planck Institute for Multidisciplinary Sciences, 37075 Göttingen, Germany, Collaborative Research Center 889, University of Göttingen, 37075 Göttingen, Germany, Multiscale Bioimaging Cluster of Excellence (MBExC), University of Göttingen, 37075 Göttingen, Germany

For correspondence: tpangrs@gwdg.de

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Editors

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Reviewer #1 (Public Review):

Summary:

This manuscript dissects the contribution of the CaBP 1 and 2 on the calcium current in the cochlear inner hair cells. The authors measured the calcium current inactivation from the double knock-out CaBP1 and 2 and showed that both proteins contribute to voltage-dependent and calcium-dependent inactivation. Synaptic release was reduced in the double KO. As a consequence, the authors observed a depressed activity within the auditory nerve. Taken together, this study identifies a new player that regulates the stimulation-secretion coupling in the auditory sensory cells.

Strengths:

In this study, the authors bring compelling evidence that CaBP 1 and 2 are both involved in the inactivation of the calcium current, from cellular up to system level, and by taking care to probe different experimental conditions such as different holding potentials and by rescuing the phenotype with the re-expression of CaBP2. Indeed, while changing the holding potential worsens the secretion, it completely changes the kinetics of the inactivation recovery. It alerts the reader that probing different experimental conditions that may be closer to physiology is better suited to uncovering any deleterious phenotype. This gave pretty solid results.

Weaknesses:

Although this study clearly points out that CaBP1 is involved in the calcium current inactivation, it is not clear how CaBP1 and CaBP2 act together (but this is probably beyond the scope of the study). Another point is that the authors re-express CaBP2 to largely rescue the phenotype in the double KO but no data are available to know whether the re-expression of both CaBP1 and CaBP2 would achieve a full recovery and what would be the effect of the sole re-expression of CaBP1 in the double KO.

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Reviewer #2 (Public Review):

Summary:

In the manuscript by Oestreicher et al, the authors use patch-clamp electrophysiology, immunofluorescent imaging of the cochlea, auditory function tests, and single-unit recordings of auditory afferent neurons to probe the unique properties of calcium signaling in cochlear hair cells that allow rapid and sustained neurotransmitter release. The calcium-binding proteins (CaBPs) are thought to modify the inactivation of the Cav1.3 calcium channels in IHCs that initiate vesicle fusion, reducing the calcium-dependent inactivation (CDI) of the channels to allow sustained calcium influx to support neurotransmitter release. The authors use knockout mice of *Cabp1* and *Cabp2* in a double knockout (*Cabp1/2* DKO) to show that these molecules are required for enabling sustained calcium currents by reducing CDI and enabling proper IHC neurotransmitter release. They further support their evidence

by re-introducing *Cabp2* using an injection of AAV containing the *Cabp2* sequence into the cochlea, which restores some of the auditory function and reduces CDI in patch-clamp recordings.

Strengths:

Overall the data is convincing that *Cabp1/2* is required for reducing CDI in cochlear hair cells, allowing their sustained neurotransmitter release and sound encoding. Figures are well-prepared, recordings are careful and stats are appropriate, and the manuscript is well-written. The discussion appropriately considers aspects of the data that are not yet explained and await further experimentation.

Weaknesses:

There are some sections of the manuscript that pool data from different experiments with slightly different conditions (wt data from a previous paper, different calcium concentrations, different holding voltages, tones vs clicks, etc). This makes the work harder to follow and more complicated to explain. However, the major conclusion, that *cabp1* and *2* work together to reduce calcium-dependent inactivation of L-type calcium channels in cochlear inner hair cells, still holds.

Another weakness is that the authors used injections of AAV-containing sequences for *Cabp2*, but do not present data from sham surgeries. In most cases, the improvement of hearing function with AAV injection is believable and should be attributed to the *cabp2* function. However, in at least one instance (Figure 4B), the results of the AAV injection experiments may be overinterpreted - the authors show that upon AAV injection, the hair cells have a much longer calcium current recovery following a large, long depolarization to inactivate the calcium channels. Without comparison to sham surgery, it is not known if this result could be a subtle result of the surgery or indeed due to the *Cabp2* expression.

It would be great to see the auditory nerve recordings in AAV-injected animals that have a recovery of ABRs. However, this is a challenging experiment that requires considerable time and resources, so is not required.

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Reviewer #3 (Public Review):

Summary:

The authors attempted to unravel the role of the Ca^{2+} -binding proteins CaBP1 and CaBP2 for the hitherto enigmatic lack of Ca^{2+} -dependent inactivation of Ca^{2+} currents in sensory inner hair cells (IHCs). As Ca^{2+} currents through Cav1.3 channels are crucial for exocytosis, the lack of inactivation of those Ca^{2+} currents is essential for the indefatigable sound encoding by IHCs. Using a deaf mouse model lacking both CaBP1 and CaBP2, the authors convincingly demonstrate that both CaBP1 and CaBP2 together confer a lack of inactivation, with CaBP2 being far more effective. This is surprising given the mild phenotype of the single knockouts, which has been published by the authors before. Re-admission of CaBP2 through viral gene transfer into the inner ear of double-knockout mice largely restored hearing function, normal Ca^{2+} current properties, and exocytosis.

Strengths:

1. In vitro electrophysiology: perforated patch-clamp recordings of $\text{Ca}^{2+}/\text{Ba}^{2+}$ currents of inner hair cells (IHCs) from 3-4 week-old mice - very difficult recordings - necessary to not interfere with intracellular Ca^{2+} buffers, including CaBP1 and CaBP2.
2. Capacitance (exocytosis) recordings from IHCs in perforated patch mode.
3. The insight that a negative holding potential might underestimate the impact of lack of CaBP1/2 on the inactivation of I_{Ca} in IHCs. As the physiological holding potential is much more positive than a preferred holding potential in patch clamp experiments it has a strong

impact on inactivation in the pauses between depolarization mimicking receptor potentials. This truly advances our thinking about the stimulation of IHCs and accumulating inactivation of the Cav1.3 channels.

4. Insight that the voltage sine method with usual voltage excursions (35 mV) to determine the membrane capacitance (for exocytosis measurements) also favors the inactivated state of Cav1.3 channels
5. Use of double ko mice (for both CaBP1 and CaBP2, DKO) and use of DKO with virally injected CaBP2-eGFP into the inner ear.
6. Use of DKO animals/IHCs/SGNs after virus-mediated CaBP2 gene transfer shows a great amount of rescue of the normal ICa inactivation phenotype.
7. In vivo measurements of SGN AP responses to sound, which is highly demanding.
8. In vivo measurements of hearing thresholds, DPOAE characteristics, and ABR wave I amplitudes/latencies of DKO mice and DKO+injected mice compared to WT mice.

Very thorough analysis and presentation of the data, excellent statistical analysis.

The authors achieved their aims. Their results fully support their conclusions. The methods used by the authors are state-of-the-art.

The impacts on the field are the following:

Regulation of inactivation of Cav1.3 currents is crucial for the persistent functioning of Cav1.3 channels in sensory transduction.

The findings of the authors better explain the phenotype of the human autosomal recessive DFNB93, which is based on the malfunction of CaBP2.

Future work - by the authors or others - should address the molecular mechanisms of the interaction of CaBP1 and 2 in regulating Cav1.3 inactivation.

Weaknesses:

I do not see weaknesses.

What is not explained (but was not the aim of the authors) is how the CaBPs 1 and 2 interact with the Cav1.3 channels and with each other to reduce CDI. Also, why DFNB93, which is based on mutation of the CaBP2 gene, lead to a severe phenotype in humans in contrast to the phenotype of the CaBP2 ko mouse.

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